

1.3 - StorageCompression - ProgTask.pdf

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IGCSE Computer Science

1.1 Data Representation

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1.3 – Dictionary Decoder

Write a program that accepts a single HEX string as input.

Each hex digit corresponds to an item in the dictionary below:

0	TH	4	NO	8	C	C	F
1	IS	5	USE	9	.	D	ER
2	A	6	T	A	SPACE	E	RY
3	WAY	7	DO	B	E	F	OR

Your program should decode the hex string entered and output the correctly formed message.

An appropriate error message should be displayed for invalid input.

Test your program using the following inputs:

- 5a0bacf8b9
- 01a1a0ba39
- 0dba1a240d999
- 7AFA7A469A0DBA1A4A6E9
- T41515N0T0KAY (Test for invalid input)

If you have coded it correctly, your output should make sense!

Top Tips:

- Use an indexed data structure for your dictionary
- Use a loop to iterate through each character in turn